

Project Summary: HealthConnect Appointment No-Show Prediction

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1. Specific Problem

The core business problem I am addressing is the operational inefficiency and lost revenue caused by high rates of patient "No-Shows" at HealthConnect Clinic. My objective as a Data Scientist is to define a machine learning problem that predicts which patients are likely to miss their scheduled appointments. By identifying these high-risk patients at the time of booking, the clinic can deploy proactive interventions—such as targeted SMS reminders or smart overbooking strategies—to optimize clinical schedules and improve patient care.

2. Resources Used

To assess the viability of this machine learning solution, I utilized two primary resources provided by HealthConnect: HealthConnect Appointment Data (HealthConnect_Appointment_Data.csv): A historical dataset containing 5,000 appointment records and 18 variables detailing patient demographics, booking logistics, and historical reliability.

HealthConnect Data Dictionary: Used to understand the precise definitions, data types, and business context of each variable, particularly regarding missing values and derived features.

3. Key Observations

During my initial data assessment, I identified several critical factors that will shape my modelling approach:

Outcome Distribution: The dataset contains three distinct outcomes: Attended, No-Show, and Cancelled. "No-Shows" represent the minority class, indicating that I will face a class imbalance challenge during model training.

Data Quality Issues: Variables such as `distance_to_clinic_km` and `waiting_time_minutes` contain limited missing values that will require imputation strategies.

Data Leakage Risk: I identified a severe data leakage risk with the `waiting_time_minutes` variable. If this represents the actual time a patient spent in the waiting room, this value is only generated if the patient actually shows up. Using it to predict attendance would result in an artificially perfect model that fails in production.

4. Proposed Approach

Based on my analysis, I have defined the following machine learning approach:

Target Variable Formulation: I will frame this as a Binary Classification problem. I made the strategic decision to filter out all "Cancelled" appointments. A cancellation is an active communication from the patient that frees up the clinic's schedule, whereas a "No-Show" is the passive, unpredictable behaviour I want to prevent.

Binarization: The remaining target variable (appointment_outcome) will be encoded as 1 for No-Show (the positive class I want to predict) and 0 for Attended (the negative baseline).

Algorithm Selection: I plan to establish a baseline using Logistic Regression (for interpretability) and then deploy advanced tree-based ensemble models like Random Forest and XGBoost, which excel at capturing non-linear relationships in tabular healthcare data.

5. Key Considerations

As I move toward model development, I am keeping three major considerations in mind:

Handling Class Imbalance: Because "No-Shows" are the minority class, I must implement techniques such as SMOTE (Synthetic Minority Over-sampling Technique) or class weighting. Otherwise, the model will simply predict "Attended" for every patient and achieve high accuracy but zero business value.

Evaluation Metrics: Standard Accuracy will be misleading. I will prioritize Recall (to ensure I catch as many potential No-Shows as possible), the F1-Score, and the ROC-AUC curve to evaluate my model's true predictive power.

Ethical Bias: I must ensure my model evaluates behavioural and logistical features (like lead time and past no-shows) rather than relying heavily on protected demographic attributes (like age or gender) to avoid unfairly penalizing specific patient groups.

6. Proposed Focus for Week 5

In Week 5, I will transition from problem definition to Machine Learning Model Development and Evaluation. My focus will be on executing the data pre-processing pipeline (handling missing values, encoding categorical variables, and dropping the leaking waiting_time_minutes feature). I will then split the data, train my classification models, apply class imbalance corrections, and rigorously evaluate their performance to deliver actionable business recommendations to HealthConnect.